

Column-Aware Reading Order for Scholarly Documents

Bea Placeholder

Cyril Sample
Institute for Illustrative Research

Abstract

Readers of scholarly documents expect text to flow down the left column and then down the right one, with titles, abstracts and wide figures interrupting that flow where they sit on the page. Software that extracts text often gets this wrong. We describe a simple model of column structure, explain how wide elements are placed within it, and report how often a naive top-to-bottom order goes astray on a small collection of papers.

1 Introduction

A two-column page is read as two narrow pages placed side by side. The eye travels down the left column, returns to the top of the right column, and continues from there. Anything that spans both columns, such as the title block above, is read at the point where it appears, before the columns that follow it.

Extraction tools that sort text purely by vertical position interleave the two columns line by line, producing sentences that jump back and forth between unrelated paragraphs.¹ The problem is easy to state but surprisingly easy to get wrong in practice, because real pages mix columns, wide floats, footnotes and running headers.

2 Related Work

Document layout analysis has a long history in the processing of scanned office documents, where the input is a raster image and the regions of text must be found before they can be read. Born-digital documents make the first step easier, since the characters and their positions are known exactly, but the grouping of characters into lines, lines into blocks and blocks into a reading sequence remains the same problem.

2.1 Layout analysis

Most methods look for a vertical band of white space that separates the columns over a large part of the page. Wide elements break that band, and a robust method has to treat the page as a stack of horizontal regions, each with its own number of columns, rather than as a single grid.

¹This failure is common enough that many readers have learned to distrust copied text from such papers.

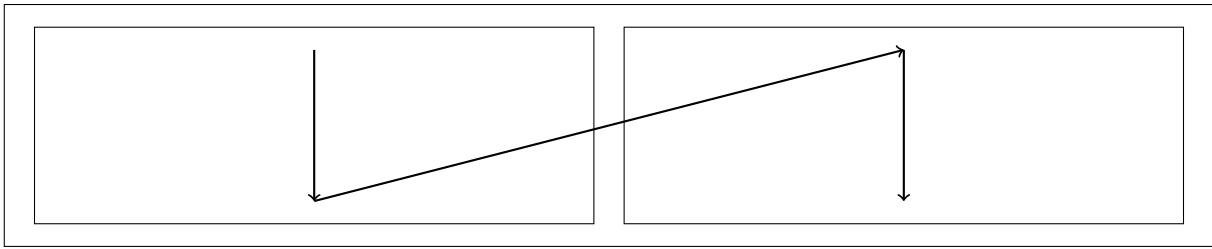


Figure 1: The reading path on a two-column page runs down the left column, back up to the top of the right column, and down again. Wide elements such as this figure are read where they sit, before the columns below.

3 Approach

We first group text items into lines by their baselines, then split each line where it crosses the gap between the columns. Regions are formed from runs of consecutive lines that share the same split.

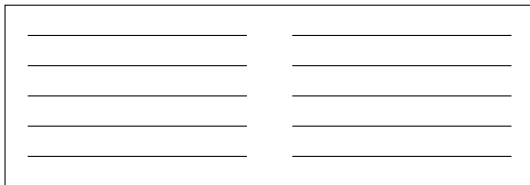


Figure 2: Lines are split at the gap between the columns before regions are formed.

Within a region the left column is emitted before the right one. Regions themselves are emitted from the top of the page to the bottom, so a wide figure that sits between two bands of columns is read between them.

3.1 Handling floats

Figures and tables are treated as blocks rather than as text. A caption is attached to the float it describes, so that it is read together with the figure and not as part of the surrounding paragraph.

3.2 Footnotes and page furniture

Footnotes are read at the end of the column in which they appear. Running headers and page numbers are excluded from the reading sequence entirely, since they repeat on every page and carry no part of the argument.

4 Evaluation

We checked the model against a handful of papers with hand-labelled reading order. A naive order that sorts lines by their vertical position was wrong on every two-column page, while the region model was wrong only where a caption had been set in the margin.

Most remaining errors involved short lines, such as equation numbers and section headings, whose horizontal position did not clearly place them in either column.

5 Conclusion

Treating a page as a stack of regions with their own column counts handles the common cases of scholarly layout with very little machinery.

References

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- [3] M. Moe. *A Primer on Page Segmentation*. Example Press, 2015.